

In-class Quiz

- ① length: 5
symmetric
 $\omega_1 = 0.1$
 $\omega_2 = 0.4$
 $\omega_3 = 0.7$

Design coef so filter only passes mid frequency ie ω_2 .

$$h[n] = \{h[0], h[1], h[2], h[3], h[4]\}$$

$$\alpha = h[0] = h[4]$$

$$\beta = h[1] = h[3]$$

$$\gamma = h[2]$$

$$Y(e^{j\omega}) = \alpha X(e^{j\omega}) + \beta X(e^{j\omega})e^{-j\omega} + \gamma X(e^{j\omega})e^{-j2\omega} + \beta X(e^{j\omega})e^{-j3\omega} + \alpha X(e^{j\omega})e^{-j4\omega}$$

$$\hookrightarrow H(e^{j\omega}) = \alpha + \beta e^{-j\omega} + \gamma e^{-j2\omega} + \beta e^{-j3\omega} + \alpha e^{-j4\omega}$$

$$= \alpha(1 + e^{-j4\omega}) + \beta(e^{-j\omega} + e^{-j3\omega}) + \gamma e^{-j2\omega}$$

$$|H(e^{j0.1})| = 0$$

$$|H(e^{j0.4})| = 1$$

$$|H(e^{j0.7})| = 0$$

$$|H(e^{j\omega})| = 2\alpha \cos(2\omega) + 2\beta \cos(\omega) + \gamma$$

$$\angle H(e^{j\omega}) = -2\omega \quad H(e^{j\omega}) = (2\alpha \cos(2\omega) + 2\beta \cos(\omega) + \gamma) e^{-j2\omega}$$

$$\hookrightarrow 2 \cos(2(0.1)) \alpha + 2 \cos(0.1) \beta + \gamma = 0$$

$$2 \cos(2(0.4)) \alpha + 2 \cos(0.4) \beta + \gamma = 1$$

$$2 \cos(2(0.7)) \alpha + 2 \cos(0.7) \beta + \gamma = 0$$

$$\alpha = h[0] = h[4] = -21.64$$

$$\beta = h[1] = h[3] = 76.18$$

$$\gamma = h[2] = -109.17$$

- ② length: 5
symmetric

same eqn as ①

$$|H(e^{j0})| = 0$$

$$\hookrightarrow |H(e^{j\omega})| = 2\alpha \cos(2\omega) + 2\beta \cos(\omega) + \gamma$$

$$|H(e^{j\pi/2})| = 1$$

$$|H(e^{j\pi})| = 0$$

$$\begin{aligned} 2\alpha + 2\beta + \gamma &= 0 \\ -2\alpha + 0 + \gamma &= 1 \\ 2\alpha - 2\beta + \gamma &= 0 \end{aligned}$$

$$\begin{aligned} \alpha = h[0] = h[4] &= -0.25 \\ \beta = h[1] = h[3] &= 0 \\ \gamma = h[2] &= 0.5 \end{aligned}$$